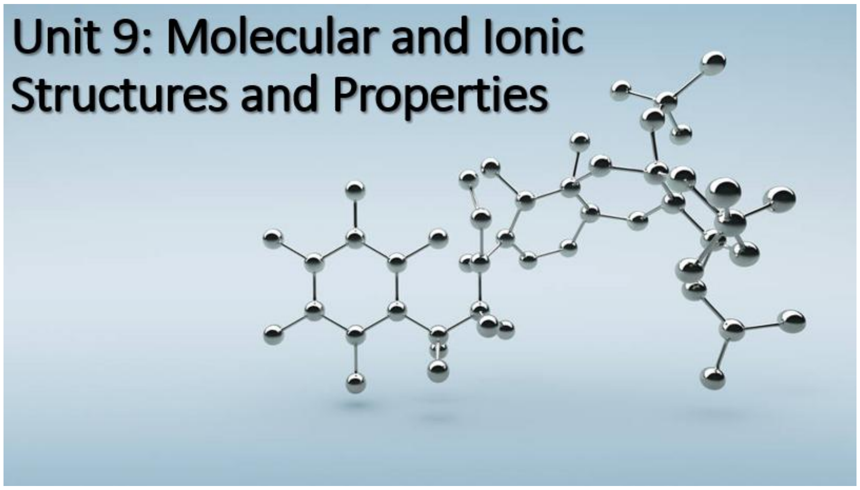
Name: ____ Period: ____________

Unit 9: Compound Structure and Properties

AP Classroom Topic 2

Unit Map



Textbook Reference: Tro Chemistry: A Molecular Approach Chapter 9-10

Date	Day	Learning Target	Activities	Work on this
3/11	U9D1	How are atoms bonded together?	Self Study: Watch Lectures 1-4 BEFORE the end of Spring Break	Lectures 1-4- quiz on 3/24-25
3/12-13	U9D2	What is a Lewis Dot Structure?	~U8 Mini Test Self Study: Watch Lectures 1-4 BEFORE the end of Spring Break	Lectures 1-4- quiz on 3/23-24
3/23-24	U9D3	How can we predict the shape of a molecule?	~Open note quiz! Lectures 1-4	
3/25-26	U9D4	What are hybrid orbitals?		
3/27-30	U10D1	What is an intermolecular force?	Review Bonding Begin Unit 10: IMFs	
3/31-4/1	U9D5	What have you learned about bonding?	Unit 8 Miniexam	

	Unit 9: Compound Structure and Properties
2.1	**Types of Chemical Bonds** Explain the relationship between the type of bonding and the properties of the elements participating in the bond.
2.1A	2.1.A.1: Electronegativity values for the representative elements increase going from left to right across a period and decrease going down a group. These trends can be understood qualitatively through the electronic structure of the atoms, the shell model, and Coulomb’s law.
	2.1.A.2: Valence electrons shared between atoms of similar electronegativity constitute a nonpolar covalent bond. For example, bonds between carbon and hydrogen are effectively nonpolar even though carbon is slightly more electronegative than hydrogen.
	2.1.A.3: Valence electrons shared between atoms of unequal electronegativity constitute a polar covalent bond. The atom with a higher electronegativity will develop a partial negative charge relative to the other atom in the bond. In single bonds, greater differences in electronegativity lead to greater bond dipoles. All polar bonds have some ionic character, and the difference between ionic and covalent bonding is not distinct but rather a continuum.
	2.1.A.4: The difference in electronegativity is not the only factor in determining if a bond should be designated as ionic or covalent. Generally, bonds between a metal and nonmetal are ionic, and bonds between two nonmetals are covalent. Examination of the properties of a compound is the best way to characterize the type of bonding.
	2.1.A.5: In a metallic solid, the valence electrons from the metal atoms are considered to be delocalized and not associated with any individual atom.
2.2	**Intramolecular Force and Potential Energy** Represent the relationship between potential energy and distance between atoms, based on factors that influence the interaction strength.
2.2A	2.2.A.1: A graph of potential energy versus the distance between atoms (internuclear distance) is a useful representation for describing the interactions between atoms. Such graphs illustrate both the equilibrium bond length (the separation between atoms at which the potential energy is lowest) and the bond energy (the energy required to separate the atoms).
	2.2.A.2: In a covalent bond, the bond length is influenced by both the size of the atom’s core and the bond order (i.e., single, double, triple). Bonds with a higher order are shorter and have larger bond energies.
	2.2.A.3: Coulomb’s law can be used to understand the strength of interactions between cations and anions. i. Because the interaction strength is proportional to the charge on each ion, larger charges lead to stronger interactions. ii. Because the interaction strength increases as the distance between the centers of the ions (nuclei) decreases, smaller ions lead to stronger interactions.
2.3	**Structure of Ionic Compounds** Represent an ionic solid with a particulate model that is consistent with Coulomb’s law and the properties of the constituent ions.
2.3A	2.3.A.1: The cations and anions in an ionic crystal are arranged in a systematic, periodic 3-D array that maximizes the attractive forces among cations and anions while minimizing the repulsive forces.
2.4	**Structure of Metals and Alloys** Represent a metallic solid and / or alloy using a model to show essential characteristics of the structure and interactions present in the substance.
2.4A	2.4.A.1: Metallic bonding can be represented as an array of positive metal ions surrounded by delocalized valence electrons (i.e., a “sea of electrons”).
	2.4.A.2: Interstitial alloys form between atoms of significantly different radii, where the smaller atoms fill the interstitial spaces between the larger atoms (e.g., with steel in which carbon occupies the interstices in iron).
	2.4.A.3: Substitutional alloys form between atoms of comparable radius, where one atom substitutes for the other in the lattice. (e.g, in certain brass alloys, other elements, usually zinc, substitute for copper.)
2.5	**Lewis Diagrams** Represent a molecule with a Lewis diagram.
2.5A	2.5.A.1: Lewis diagrams can be constructed according to an established set of principles.
2.6	**Resonance and Formal Charge** Represent a molecule with a Lewis diagram that accounts for resonance between equivalent structures or that uses formal charge to select between nonequivalent structures.
2.6A	2.6.A.1: In cases where more than one equivalent Lewis structure can be constructed, resonance must be included as a refinement to the Lewis structure. In many such cases, this refinement is needed to provide qualitatively accurate predictions of molecular structure and properties.
	2.6.A.2: The octet rule and formal charge can be used as criteria for determining which of several possible valid Lewis diagrams provides the best model for predicting molecular structure and properties.
	2.6.A.3: As with any model, there are limitations to the use of the Lewis structure model, particularly in cases with an odd number of valence electrons.
2.7	**VSEPR and Hybridization** Based on the relationship between Lewis diagrams, VSEPR theory, bond orders, and bond polarities: i. Explain structural properties of molecules. ii. Explain electron properties of molecules.
2.7A	2.7.A.1: VSEPR theory uses the Coulombic repulsion between electrons as a basis for predicting the arrangement of electron pairs around a central atom.
	2.7.A.2: Both Lewis diagrams and VSEPR theory must be used for predicting electronic and structural properties of many covalently bonded molecules and polyatomic ions, including the following:
	Molecular geometry (linear, trigonal planar, tetrahedral, trigonal pyramidal, bent, trigonal bipyramidal, seesaw, T-shaped, octahedral, square pyramidal, square planar) Bond angles Relative bond energies based on bond order Relative bond lengths (multiple bonds, effects of atomic radius) Presence of a dipole moment
	Hybridization of valence orbitals for atoms within a molecule or polyatomic ion
	2.7.A.3: The terms “hybridization” and “hybrid atomic orbital” are used to describe the arrangement of electrons around a central atom. When the central atom is sp hybridized, its ideal bond angles are 180°; for sp2 hybridized atoms the bond angles are 120°; and for sp3 hybridized atoms the bond angles are 109.5°. ★ ★
	2.7.A.4: Bond formation is associated with overlap between atomic orbitals. In multiple bonds, such overlap leads to the formation of both sigma and pi bonds. The overlap is stronger in sigma than pi bonds, which is reflected in sigma bonds having greater bond energy than pi bonds. The presence of a pi bond also prevents the rotation of the bond and leads to geometric isomers. ★

Molecular and Ionic Structure and Properties

Section 1 – Lewis Symbols and the Octet Rule

(a) Complete the Lewis electron-dot symbols for each of the following elements by drawing the valence electrons in an appropriate manner.

Na Mg Al Si P S Cl Ar

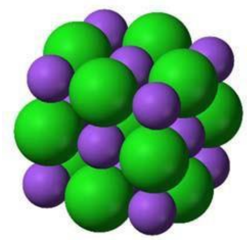
(b) Write a definition of the octet rule. Try to avoid using the word "eight" in your definition.

(c) There are several types of bonding found in substances. All have to do with how ______________ ______________________________ behave.

- Covalent- valence electrons are _________________
- Polar covalent- valence electrons are _______________ __________________
- Ionic- valence electrons are ______________________
- Metallic- valence electrons are ___________________

Section 2 – Ionic Bonding

(a) This is a diagram of the crystal lattice of NaCl.



How can you identify which ion is Na^+ and which ion is Cl^-?

(b) Fill in the valence electrons on both sides of the chemical equation shown below. Draw an arrow to indicate the direction of electron transfer in this reaction.
Draw the appropriate charges on the ions in the product.

$$\mathrm{Na} \; + \; \mathrm{Cl} \; \rightarrow \; \mathrm{Na} \;\; \mathrm{Cl}$$

(c) Describe several properties of ionic substances.

(d) When sodium loses an electron, this is an (endothermic exothermic) process.

When chlorine gains an electron, this is an (endothermic exothermic) process.

$$Na(g) \rightarrow Na^+(g) + e^- \qquad \Delta H = +496 \text{ kJ}$$

$$Cl(g) + e^- \rightarrow Cl^-(g) \qquad \Delta H = -349 \text{ kJ}$$

(e) If we combine the two reactions shown above, we might think that the formation of sodium chloride is an endothermic process. Why is this view incorrect?

(f) The principal reason ionic compounds are stable is the ______________________________________

____. This can be examined with Coulomb's law.

(g) Define lattice energy.

(h) Label each of the following reactions with a ΔH value of either +788 kJ or –788 kJ.

$$NaCl(s) \rightarrow Na^+(g) + Cl^-(g) \qquad \Delta H =$$

$$Na^+(g) + Cl^-(g) \rightarrow NaCl(s) \qquad \Delta H =$$

(i) What is Coulomb's Law?

For a given arrangement of ions, the lattice energy increases as the _____________________________________

____________________increase and as their________________________decrease.

(j) why MgO has a higher lattice energy than NaF.

(k) Explain why NaF has a higher lattice energy than KCl.

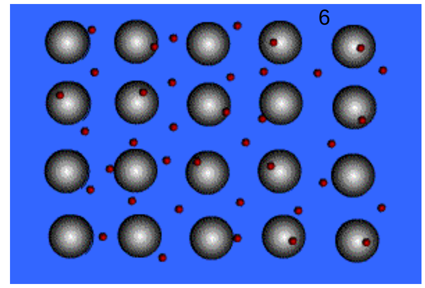
(l) Using your knowledge of Periodic Trends, arrange these three compounds in order of increasinglattice energy: LiCl, KBr, and $MgCl_2$. Justify your answer.

(m) Even though lattice energy increases with increasing ionic charge, we never find ionic compounds that contain Na^{2+} ions. Explain.

Consider how the lattice energy affect properties of compounds?:

- Melting point? Larger Lattice energy (raises lowers) the melting point.
- Boiling point? Larger Lattice energy (raises lowers) the boiling point.
- Solubility? Larger Lattice energy (raises lowers) the solubility.

The 3 ionic compounds analyzed in the lab were: NaCl, $CaCl_2$ and $CaCO_3$. Which one was not soluble? Justify your answer using Coulomb's Law



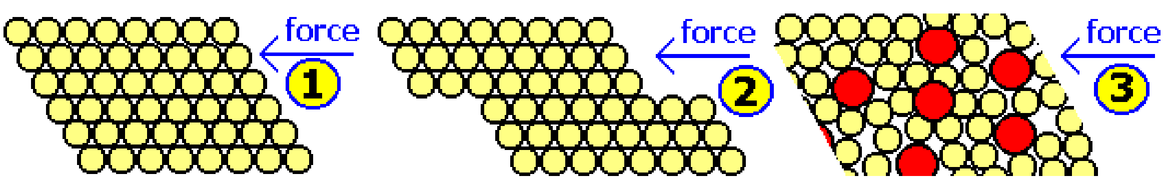
Section 3: Metallic Bonds

a) Nature of metallic bonds

- Formed between atoms of ____________________metallic elements
- Electron cloud around atoms flowing freely- often called the ____________________________unlike ionic or covalent, valence electrons are not localized
- Good conductors at all states, lustrous, very high melting points, not soluble, mostly solid
- Examples; Na, Fe, Al, Au, Co

b) Metals Form Alloys

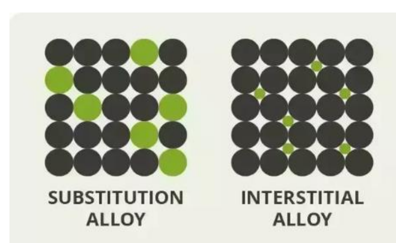
- Metals do not _____________with other metals. They form homogeneous solutions called _________________. which are a ________________of one metal in another metal.
- Examples are steel, brass, bronze and pewter.
- They can make a metal structure more resistant to ___________________________________.



c) There are two types of alloys:

a. __________________- where one metal atom type substitutes in the lattice placement of the other metal atom

b. __________________- where one metal atom type settles in the spaces within the lattice of the other metal atom

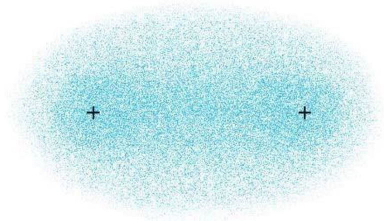


c. Classify the two alloys as substitutional or interstitial:

Brass (copper and zinc solution)...... substitutional or interstitial

Steel (iron and carbon solution)....... substitutional or interstitial

Section 4 – Covalent Bonding



(a) The diagram at right represents the electron distribution in a molecule of hydrogen. Label the region on the diagram that represents the electrons involved in covalent bonding.

(b) The atoms in the H_2 molecule are held together because the two ________________________________

are attracted to the concentration of negative charge between them.

(c) In writing Lewis structures, we usually show each______________________electron pair as a line,

and any________________________electron pairs as dots.

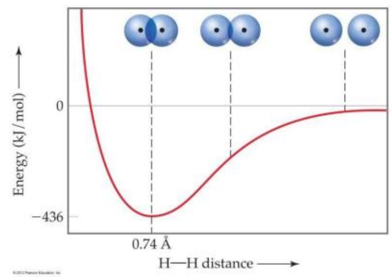
(d) Write the Lewis structure for the diatomic molecules F_2, O_2, and N_2

(e) As a general rule, the length of the bond between two atoms________________________as the number of shared electron pairs increases.

Section 5 – Covalent Bonding and Orbital Overlap

(a) The___________________of orbitals allows two electrons of opposite spin to share the space

between the nuclei, forming a ________________________________



Refer to the diagram at right.

(b) When the two H atoms are very far apart, the potential energy approaches

what value? _______________________

(c) As the two H atoms move closer together, the potential energy ________________as the strength of the bond

______________________.

(d) When the distance between the two H atoms becomes less than 0.74 Å, the potential energy

______________________because of the ____

(e) The observed_________________________is the distance at which the _________________________

forces between unlike charges (electrons and nuclei) are balanced by the _________________________

forces between like charges (electron-electron and nucleus-nucleus)

Section 6 – Strengths of Covalent Bonds

(a) The bond enthalpy is the energy required to ____
____

(b) The greater the bond enthalpy, the___________________________the bond.

(c) Breaking a bond is always an___________________________process, and forming a bond is always an_____________________________process.

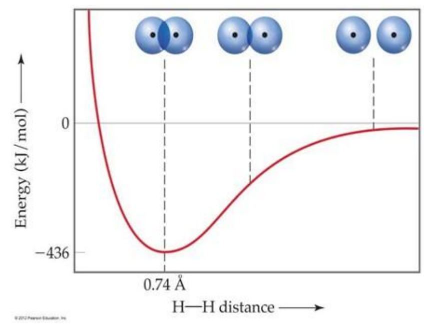
(d) In general, as the number of bonds between two atoms increases, the bond grows ______________________________ and_________________________. This trend is illustrated in table 9.4 for N–N single, double, and triple bonds.

Rank from weakest to strongest:

double bond triple bond single bond

Rank from shortest to longest:

double bond triple bond single bond



Section 7 – Bond Polarity and Electronegativity

(a) Bond polarity is a measure of how equally or unequally the electrons in any covalent bond are shared. Give two examples of each of the following bond types.

nonpolar covalent bond ______ _______

polar covalent bond ______ _______

ionic bond ______ _______

(b) Define electronegativity.

(c) If an atom has a large negative value for its electron affinity and a rather high value for its 1st ionization energy, it should have a rather____________value for electronegativity.

(d) The periodic trend is that electronegativity tends to________________________from left to right across a period, and it tends to___________________________from top to bottom down a group.

(e) Use electronegativity values to explain why the F–F bond is nonpolar and the H–F bond is polar.

(f) Each of the following bonds is polar. Use the periodic trends in electronegativity to help you to label each atom with either a partial positive charge ($\delta+$) or a partial negative charge ($\delta-$).

H—O N—C I—F Si—Br Cl—As

(g) The greater the difference in electronegativity between two atoms, the ______________________________ their bond is.

(h) If we use an arrow with a crossed end like this ($+\!\!\longrightarrow$) to denote the charge separation in a polar bond, the arrow always points toward the atom that has a __________electronegativity value.

(i) Whenever two electrical charges of equal magnitude but opposite sign are separated by a distance, a _ ____________________is established.

(j) The quantitative measure of the magnitude of a dipole is called its _________________________________

(k) If a molecule is nonpolar, it has a dipole moment of______________________.

(l) Even though there is a continuum between the extremes of ionic and covalent bonding, the simplest approach is to assume that the bond between a metal and a nonmetal is _______and that the bond between two nonmetals is_____.

Types of BONDS (not compounds)

	Ionic	Polar Covalent	Pure Covalent	Metallic
Electrons are…..	transferred	Equally shared	Unequally shared	Sea of electrons
Types of atoms in bond	Metal - Nonmetal	Nonmetal - Nonmetal	Nonmetal - Nonmetal	All one type of metal
Electro-negativity difference	$\Delta x > 1.7$	$\sim 0.4 < \Delta x < 1.7$	$\Delta x < \sim 0.4$	0 (must be same metal)
Symbols	+ and -	$\delta+$ and $\delta-$		

Section 8 – Drawing Lewis Structures

Lewis Dot Structures reflect the position of all of the electrons within a molecular structure. Electrons are always paired and are either:

____________________ Pair: shared between two atoms

____________________ Pair: unshared pair of electrons

(a) Draw Lewis electron dot structures for each of the following molecules. There should be only single bonds in each structure.

CCl_4 NF_3 OBr_2

(b) Draw Lewis electron dot structures for each of the following molecules. There may be a double or a triple bond in each structure.

$COCl_2$ C_2H_4 HCN

(c) Draw Lewis electron dot structures for each of the following polyatomic ions.

ClO_3^- NO_3^- NO_2^+

<u>Section 9 – Formal Charges</u>

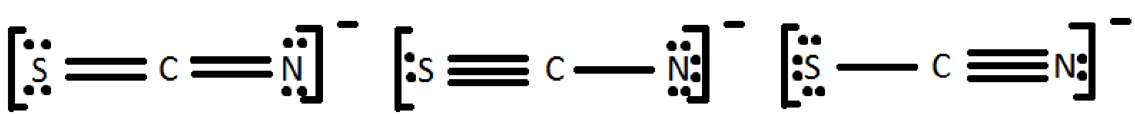
(d) The formal charge of any atom in a molecule or ion is the charge the atom would have if each bonding pair in the Lewis structure were ____

(e) Formal charge = ______________________ minus ____

(f) Calculate the formal charges on each atom in the polyatomic ions that were drawn in part (c).

(g) If more than one Lewis structure is possible, we can use formal charges to predict the dominant structure. The dominant Lewis structure is generally the one in which the atoms bear formal charges closest to_____. A Lewis structure in which any negative charges reside on the more_atoms is generally more preferred.

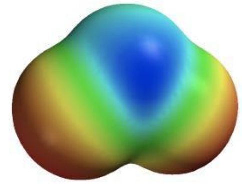
(h) Based on formal charges, which of these Lewis structures is the most dominant structure for the thiocyanate ion, SCN^-? Which is the worst structure?



(i)

Remember that formal charges do not represent real charges on atoms. These charges are just a convenient bookkeeping method that helps us decide which structure is preferred.

Section 10 – Resonance Structures



(a) Draw a LDS for ozone, O_3.

(b) Based on Figure of the actual electron density of ozone, does the molecule contain one (shorter) double bond and one (longer) single bond? Explain.

(c) When a double-headed arrow is drawn between two resonance structures, it does not mean that the two forms of the molecule are oscillating back and forth. It means that the true structure is an 'average' of the two, but not equivalent to either. Ozone has 2 ________________ '1.5' bond, **not** _____ _________________ and ______ ________________.

(d) Two or more resonance structures will have________________________arrangement of atoms, but a different ____

(e) Draw three different resonance forms for the carbonate ion.

(f) _______________________is defined as the number of bonding pairs shared between a pair of atoms. For example, the bond order in F_2 is ______, and the bond order in O_2 is ______. The bond order in N_2 is ______. Ozone has a bond order of 1.5.

Section 11 – Exceptions to the Octet Rule

(a) The Lewis structure of NO, NO_2, and ClO_2 each have Lewis structures with an ___________of electrons. Such particles are known as _____________________and are highly reactive. Draw the LDS for NO.

(b) Draw the Lewis structure of BF_3. The Lewis structure of BF_3 has a central atom with less than an octet. This structure is called ________________ because it has an incomplete octet.

(c) There are many molecules that have central atoms with more than an octet around them. Such molecules (or ions) are called hypervalent. Hypervalent molecules (or ions) are formed only for central atoms from period____and below in the periodic table. The principal reason for their formation is the __________ ____of the central atom.

(d) Circle the molecules (or ions) that should exist, based on these rules that apply to hypervalent molecules (or ions).

OF_4 SF_4 NF_5 PF_5 OF_6 SeF_6 XeF_2 IF_4^- Br_3^-

(e) Draw the Lewis structures for each of the molecules or ions that you circled in part (e).

Section 12 – Molecular Shapes

Molecular shapes are determined by the number of _______________ electron pairs around a central atom.

(a) An electron domain can consist of either a________________________pair or a ______________________ pair of electrons. Each multiple bond in a molecule also constitutes a single electron domain.

(b) The VSEPR model is based on the idea that electron domains are negatively charged and therefore one_

another. The best arrangement of a given number of electron

domains is the one that ____

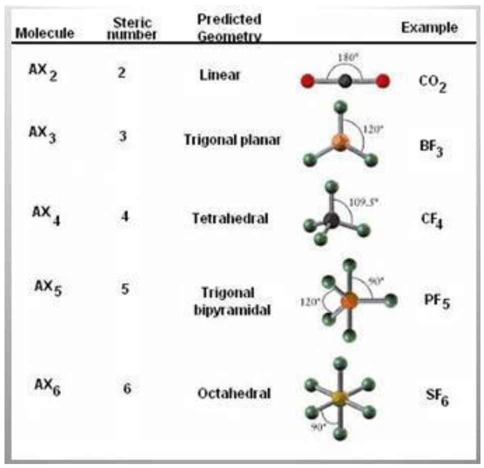
(c) What is the difference between electron domain geometry and molecular geometry? Use the water molecule as an example to explain the difference between the two.

Electron domain geometry includes all the bonding pairs and lone pairs in the shape. Molecular geometry only accounts for the bonded groups, but cannot be understood without the electron geometry.

Water has a tetrahedral shape electronically, but because the two lone pairs are invisible, the shape is 'bent' with 109.5 degree bond angles.

(d) The shape of the CO_2 molecule is _________________and has a bond angle of______.

(e) Identify the shape of the SO_2 molecule, which is not the same as that of the CO_2 molecule. Justify your answer based on the Lewis structure for SO_2.



Molecule	Steric number	Predicted Geometry		Example
AX_2	2	Linear	180°	CO_2
AX_3	3	Trigonal planar	120°	BF_3
AX_4	4	Tetrahedral	109.5°	CF_4
AX_5	5	Trigonal bipyramidal	90°, 120°	PF_5
AX_6	6	Octahedral	90°	SF_6

(f) The shape of the BF_3 molecule is_________________, and it has bond angles of____.

(g) The shape of the CH_4 molecule is_ ___________________________, and it has bond angles of_ ______________.

EXPANDED OCTET SHAPES:

(h) The shape of the PF_5 molecule is_________________________________, and it has bond angles of________________and____________.

(i) The shape of the SF_6 molecule is__________________________________, and it has bond angles of_______________.

These are all molecules with no lone pairs. Memorize these as the basis for all or the shapes.

(j) Draw the Lewis structure for each of the following molecules. Then identify the shape and the bond angles in the molecule.

carbon disulfide	arsenic pentachloride
boron trifluoride	selenium hexafluoride
silicon tetrabromide	

(a) Fill in the information in the table below. For the examples in the right side of the table, select from the following list of molecules and ions.

SO_2 SCl_2 SO_3 PCl_3 CCl_4 HCN

NO_2^- ClO_2^- NO_3^- ClO_3^- SO_4^{2-} N_3^-

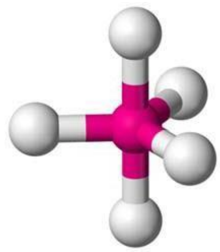
Total # Electron Domains	Electron Domain Geometry	Bonding Domains	Nonbonding Domains	Molecular Geometry	Bond Angles	Symmetric?	Molecule Example	PAI Example
2	Linear	2	0	Linear				
3	trigonal planar	3	0	Trigonal planar				
		2	1	Bent				
4	tetrahedral	4	0	Tetrahedral				
		3	1	Trigonal pyramidal				
		2	2	bent				

(b) Explain why the bond angles in NH_3 and H_2O are less than 109.5°.

(c) Explain why the Cl–C–Cl bond angle in phosgene ($Cl_2C{=}O$) is less than 120°.

(d) Molecules with five or six electron domains around the central atom have molecular geometries based on either a______________________________________(five domains) or________________________(six domains) electron domain geometry.

(e) Label the axial and the equatorial positions on the molecule shown below.



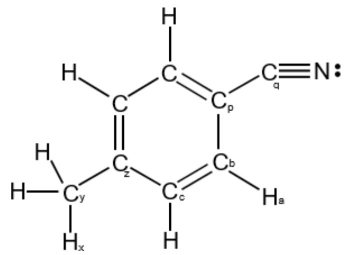
(f) Because the domains from nonbonding pairs exert_________________________repulsions than those from bonding pairs, nonbonding domains (lone pairs) always occupy the ____________________ positions in a trigonal bipyramid.

(g) Fill in the information in the table below. For the examples in the right side of the table, select from the following list of molecules and ions.

XeF_2 ClF_3 XeF_4 SF_4 ClF_5 PF_5 SF_6

Br_3^- PF_4^- IF_4^- SF_5^+ PF_6^-

Total # of Electron Domains	Electron Domain Geometry	Bonding Domains	Nonbonding Domains	Molecular Geometry	Bond Angles	Symmetry	Molecule Example	PAI Example
5	Trigonal bipyramidal	5	0	Trigonal bipyramidal				
		4	1	Seesaw				
		3	2	T shape				N/A
		2	3	Linear				
6	Octahedral	6	0	Octahedral				
		5	1	Square pyramidal				N/A
		4	2	Square planar				



(h) Based on the molecule shown above and your knowledge of the VSEPR model, predict the bond angle for each of the following bonds.

$H_x – C_y – C_z$______________ $H_a – C_b – C_c$______________ $C_p – C_q – N$ ______________

Section 13– Molecular Shape and Molecular Polarity

Draw and identify the molecular shape: CO_2 and H_2O molecules

(a) The C–O bond and the H–O bond are polar bonds because ____
____

(b) The CO_2 molecule is nonpolar because ____
____

(c) The H_2O molecule is polar because ____
____

(d) Identify the shape of each molecule and classify it as polar or nonpolar. (Don't be afraid to draw the LDS to help your prediction)

***This skill is the MOST IMPORTANT THING you can learn in the bonding unit: Identifying whether a molecule is polar or nonpolar.

Molecule	Bonding Domains	Nonbonding Domains	Molecular Geometry	Polar or Nonpolar?
CO_2				
HCN				
BF_3				
CCl_2O				
SO_2				
CH_4				
CH_3Cl				
NH_3				
H_2O				
PF_5				
PF_4Cl				
SF_4				
ClF_3				
XeF_2				
SF_6				
BrF_5				
XeF_4				

Section 14 – Hybrid Orbitals

(a) To explain molecular geometries, we can assume that the atomic orbitals on an atom (usually the central atom)______________________to form new orbitals called______________________orbitals.

(i) According to the valence-bond model, a__________________________arrangement of electron domains implies sp hybridization. This is seen in substances with two electron domains.
Ex: HCN, CO_2

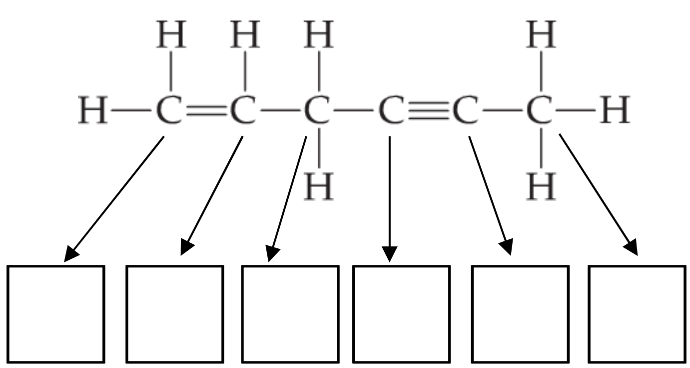
(l) According to the valence-bond model, a __________________________________arrangement of electron domains implies sp^2 hybridization. This is seen in substances with three electron domains.
Ex: H_2CO, BF_3

(l) According to the valence-bond model, a ______________________________________arrangement of electron domains implies sp^2 hybridization. This is seen in substances with 4 electron domains.
Ex: CH_4, NH_3

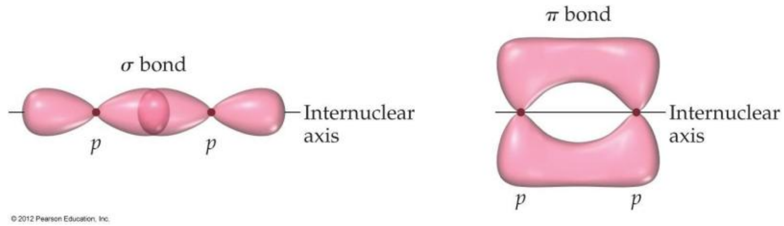
(o) Identify the orbital hybridization around the central atom in each of the following.

	# of domains	Hybridization		# of domains	Hybridization		# of domains	Hybridization
BeF_2			CCl_2O			NH_3		
BF_3			SiH_4			H_2O		
CF_4			CO_2			O_3		

(q) Identify the orbital hybridization for each carbon atom in the following molecule.



Section 15 – Multiple Bonds

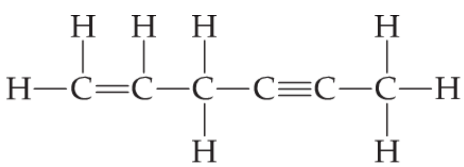


(a) A bond in which the electron density is concentrated along the line connecting the nuclei (the internuclear axis) is called a______________________________bond.

(b) To describe multiple bonding, we must consider a second kind of bond. This one is the result of the overlap between two p orbitals oriented perpendicularly to the internuclear axis. This sideways overlap of p orbitals produces a______________________________bond.

(c) In almost all cases, single bonds are_______________________bonds. A double bond consists of one________________bond and one_______________bond. A triple bond consists of one________________bond and two_______________bonds.

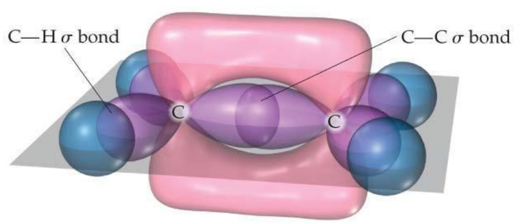
(d) How many sigma (σ) and pi (π) bonds are present in the following molecule?

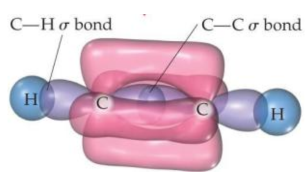


________sigma (σ) bonds

________pi (π) bonds

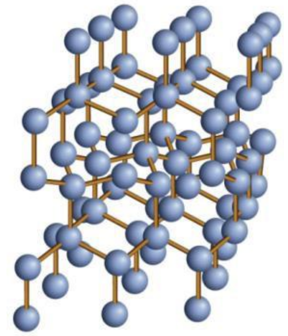
(e) Label the pi (π) bonds on the following diagrams of ethylene (C_2H_4) and acetylene (C_2H_2).





(f) For molecules that contain two or more resonance structures involving π bonds, we cannot describe the π bonds as individual bonds between neighboring atoms. Therefore we say that the π bonds are ____________________________________, or spread out among several atoms involved in the resonance structures.

Section 16: Covalent Network Solids



- Structure in which a lattice of ______________ bonds is formed- every atom is ____________________ bonded to each other
- Not dependent on ____________ to hold lattice together
- Highly organized
- Extremely ______________ structure, high mp, rigid structure
- Strongest of all bonding types
- RARE! Don’t think zebra, think horse! Examples are diamond and silicon wafers